Sumit Das

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EDUCATION

- **Khulna University of Engineering & Technology** — Khulna, Bangladesh
  B.Sc. – Electronics and Communication Engineering; CGPA: ***3.65/4.00*** — *Jan 2020 – Sept 2025*

RESEARCH INTERESTS

Research-oriented ECE graduate from KUET with research experience in **Wireless Sensor Networks (WSNs)**, integrating **Machine Learning** in network optimization for improved energy efficiency and network lifetime. Experienced in **IoT**, **Computer Vision**, and Electrical Engineering projects. Current research interests include **Wireless Communication**, **Computer Networking**, **Ad-Hoc and Vehicular Networks**, Machine Learning for network security, optimization, **Large Language Models (LLMs)**, and **Autonomous Unmanned Systems**.

SKILLS

- **Software Skills:** Cisco Packet Tracer, Ni Multisim, Proteus, MATLAB Simulink, CST Studio, Microsoft Office, Adobe Illustrator
- **Programming Languages:** Python, MATLAB, C++, HTML5, CSS3, PHP, SQL
- **Platforms:** Windows, Arduino, Raspberry Pi
- **Others:** Latex, Git, OriginPro

RESEARCH EXPERIENCE

ELITE Research Lab LLC — Queens, NY, USA
Research Intern - Remote — *Jan 2026 – Present*

- **Large Language Models:** I am currently working remotely to contribute to the lab's ongoing projects on Large Language Models and their applications.

Khulna University of Engineering & Technology — Khulna, Bangladesh
Undergraduate Researcher, Dept. of Electronics and Communication Engineering — *Feb 2024 – Sept 2025*

- **Wireless Sensor Network:** During my undergraduate thesis, I enhanced the network lifetime and energy efficiency of SEP and DEEC protocol by introducing improved probability, threshold equations and implementing advanced deployment and sleep-awake schedules of the nodes. The enhanced protocol nearly got 34% stability period improvement compared to the parent protocol.
- **Machine Learning integration with WSN:** Used Self-organizing maps for the deployment of nodes, which got around 7% better area coverage, and used clustering algorithms to organize the clusters in SEP protocol. Also used the random forest algorithm in deciding a secondary cluster-head for multi-hop routing.

PUBLICATIONS

1. **S. Das** and Md. K. Alom, "EZS-SEP: A hybrid efficient zonal stable election routing protocol with SOM-based node deployment for heterogeneous wireless sensor networks," PLOS One, vol. 21, no. 8, p. e0346479, Aug. 2026, doi: 10.1371/journal.pone.0346479.
   (**Undergraduate Thesis**)
2. S. M. Islam, **S. Das** and Md. K. Alom, "SA-DEEC: An Efficient Sleep-Awake Assisted DEEC Routing Protocol for Heterogeneous Wireless Sensor Network," (manuscript in preparation, 2026)
3. T. Tunzina, **S. Das** and MK. Morol, "Evaluating the Impact of Language Style on the Quality of VLM Outputs Using Automated Evaluation," (in progress, 2026)

PROJECTS

Robotics & Computer Vision:

i. **Target Retrieval Rover:** Built an autonomous line-following rover with a custom robotic claw. Utilized a hybrid architecture (**Arduino** & **Raspberry Pi**) with **OpenCV** for real-time color detection.

ii. **Gesture based AI Keyboard and Volume Control:** Developed a contactless human-computer interface using **Python**, **OpenCV**, and **MediaPipe**. Integrated a virtual keyboard that simulates keystrokes via thumb-index pinch gestures (using **pynput**) and a dynamic volume controller that maps finger proximity to Windows Audio API commands for real-time system interaction.

iii. **YOLO Object Detection & Tracking:** Developed a suite of real-time vision applications using **YOLOv8** and **SORT** for multi-object tracking. Implemented bi-directional traffic and pedestrian counters, and trained custom models for safety gear detection and real-time Poker Hand Analysis.

iv. **Ai Football Analysis System:** Built a **YOLO**-based analysis tool to track football players and calculate speed/distance in meters. Utilized **K-Means** for team assignment, **Optical Flow** to handle camera movement, and Perspective Transformation to accurately map the field dimensions.

Computer Networks:

i. **Campus Network:** Simulated a secure campus network (ECE, CSE, Admin) using **Cisco Packet Tracer**. Integrated DHCP, VLANs, Web/DNS servers, and RIP dynamic routing with ACL security.

Internet of Things:

i. **IoT-Based Patient Monitoring System:** Developed a system using **ESP32** and **Arduino** to track vital signs (BPM, Temp, CO2) and fall detection. Features real-time **MATLAB** visualization and automated SMS alerts.

Electronics & Simulation:

i. **Variable DC Power Supply:** Designed a 220V AC to 5V–35V regulated DC supply. Implemented short-circuit and over/under-voltage protection (> 30V, < 10V) using **LM358** op-amps and transistor logic. Integrated **LM317T** regulators, relays, and SCRs for stable output. Simulated in **Proteus** and implemented on Veroboard.

ii. **FIR Filter Design:** Designed a **MATLAB** bandpass filter (1500–2750 Hz) using Hanning/Hamming windows to suppress noise.

iii. **UWB Vivaldi Antenna Design:** Designed and simulated an Ultra-Wideband (UWB) Vivaldi Antenna using **CST Studio Suite** (2.5–4 GHz). Achieved resonance at 3.2 GHz with a return loss of -47.06 dB, VSWR ≈ 1, and 10.25 dBi directivity at 4 GHz. Optimized for narrow beamwidth (HPBW $< 70°$) to enable precise wireless detection.

Web & Database:

i. **TeliGari-BaZaar (E-commerce):** Developed an automotive e-commerce site using **PHP** and **MySQL** with secure authentication and persistent carts.

ii. **Blood Bank Management DBMS:** Created an **SQL** database to automate blood inventory, expiration tracking, and donor availability dashboards.

Explore my **Github**: SumitDas44

Test Scores

- **GRE** – **314** (Quant: **170**, Verbal: 144, AWA: 4.0)

TRAINING

EDGE Project of ICT Division KUET, Khulna, Bangladesh
Trainee, Machine Learning with Python *Dec 2024 – Feb 2025*

- Completed training on "Machine Learning with Python" under the Enhancing Digital Government and Economy (EDGE) Project of Bangladesh Computer Council, ICT Division.

Walton Hi-Tech Industries PLC Kaliakoir, Gazipur, Bangladesh
Industrial Trainee, Dept. of Production Management (Refrigerator) *April 2024*

- During this immersive experience, we delved into various production processes, including PCB manufacturing, hardware design, SMT line operations, and embedded software engineering.

Awards AND Scholarships

- **Dean's Award**, Khulna University of Engineering and Technology (KUET) – For scoring above 3.75 cgpa in the fourth year.
- **Technical Scholarship**, Khulna University of Engineering and Technology (KUET) – For good academic results from first to third year of undergraduate.
- **General Scholarship**, Dhaka Board – For outstanding result in board exams.

Extracurricular Activities

Served as General Secretary and Vice-president in different university clubs and organized numerous inter-university and national events.